

Particulate Matter Monitoring & Health Effects

Issue 2026-10-04 | Entry window 4 October 2026 (full day) | 5 records screened in scope

5

RECORDS IN SCOPE
(8 EXCLUDED)

4

OF 10 SUBTOPIC
BANDS POPULATED

0

EFFECT ESTIMATES
WITH A CI

1

TIER-A
RECORDS

0

RECORDS CARRIED ON
METADATA ALONE

Signal of the day

A Saturday of five records, and the two strongest are both about air-cleaning controls — one measured properly, one not measured at all.

Blackley and colleagues (*Ann. Work Expo. Health*, tier A) evaluated five high-volume suction systems during **51** real dental procedures, with direct-reading number and mass instruments (PM₄ respirable, PM₁₀ thoracic) beside the dentist and in the hallway. The estimand is the one that matters and is almost never used: a system counts as effective only when the **95 % lower confidence limit of the procedure-to-background ratio is ≤ 1** — that is, when the control returns the air to background *against the instrument's own noise*, not when it produces a quotable percentage reduction. Against that, Rees and colleagues report the qualitative arm of **AFRI-c**, the care-home HEPA cluster RCT that found **no reduction in respiratory infections**: units were accepted, maintenance was light, cost was judged prohibitive sector-wide — and **runtime, filter loading and achieved air-change rate were never measured**. **Why it matters for sensing**: a filtration null without exposure verification cannot distinguish “filtration does not work” from “the filters were off”. One logging low-cost node per room is the cheap repair to a trial design that otherwise cannot fail informatively.

What changed today

- **Road dust is a live microbial reservoir.** 16S metabarcoding of dust from nine sites in three Philippine metros: alpha diversity does not differ between cities (Kruskal–Wallis $p > 0.05$) but composition does (PERMANOVA $R^2 = 0.466$, $p = 0.004$), with *Paracoccus* rising toward the most metal-laden site and correlating with cobalt and nickel. Dust is a reservoir, not an exposure — nothing here measures what resuspends.
- **The atmospheric mycobiome by aircraft.** Seasonal flights joined to vegetation-productivity and meteorological data find fungal diversity declining with altitude and with productivity, decomposers and pathogens dominant, and **over 40 %** of taxa putative plant or animal pathogens. Amplicon relative abundance is not concentration and viability is untested.
- **Review, not evidence.** A *Carcinogenesis* narrative review maps early-onset lung-cancer mechanisms — mutational signatures, telomere and mitochondrial dysfunction, CXCL13/IL-1 β inflammation, PD-L1 and CD47–SIRP α immune evasion, microbiome disruption — and calls single-study biomarkers validated. Useful as a list of mediators a cohort with *measured* personal exposure could test.

Window continuity

The 3 October issue closed with `last_entry_date` at 3 October. This issue collects **4 October in full**, continuous with it. It is a Saturday, and the volume is a series low: 13 fresh candidates, **5 admitted**. Harvest and screening were completed on the 6 October run and cached; this run authored from that cache without re-querying. Dailies for 5–8 October and the W40 weekly (27 September–3 October) remain owed and follow in this backfill.

Corpus analytics

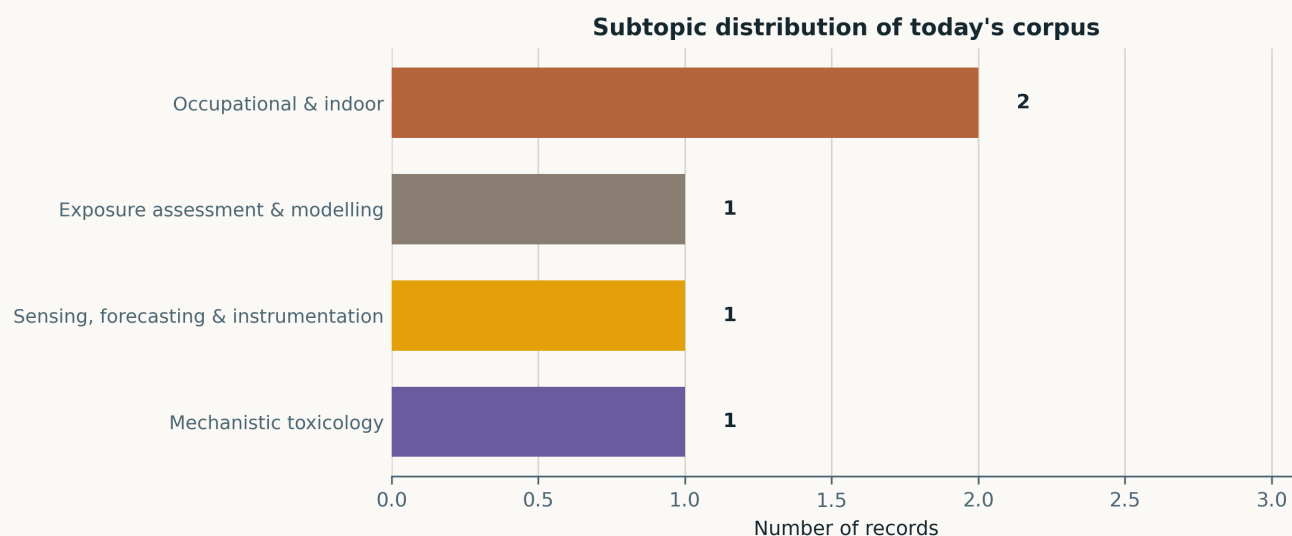


Fig. Subtopic assignment of the 5 in-scope records. **Four of ten bands populated:** *Occupational & indoor* two (both air-cleaning controls), and one each for *Exposure assessment*, *Sensing* and *Mechanistic toxicology*. No clinical-endpoint band is populated — the one cancer record is a mechanistic review, not a study of people.

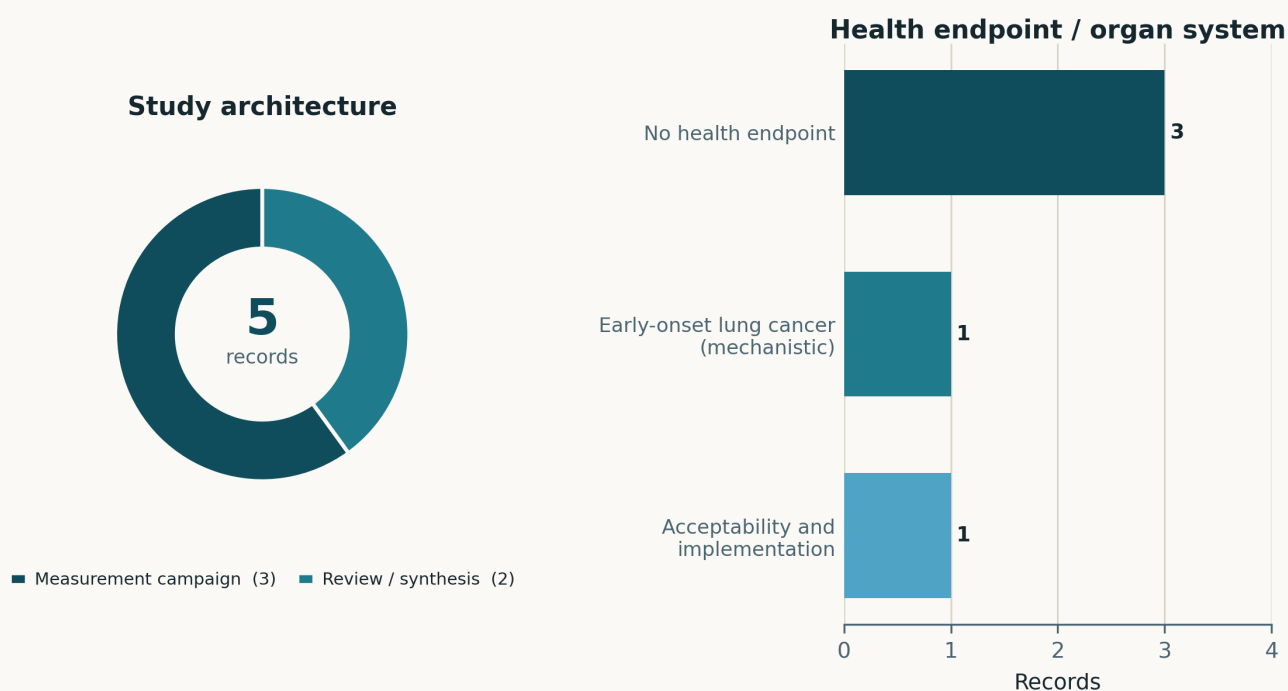


Fig. Left — architecture: **three** measurement campaigns (the dental operatories, the road-dust survey and the aircraft bioaerosol flights) and **two** filed as review / synthesis (the narrative review and the qualitative arm nested in the AFRI-c trial). Right — **all five** records report no measured health endpoint, including the HEPA record, whose trial endpoint lives in the parent RCT and not in this paper.

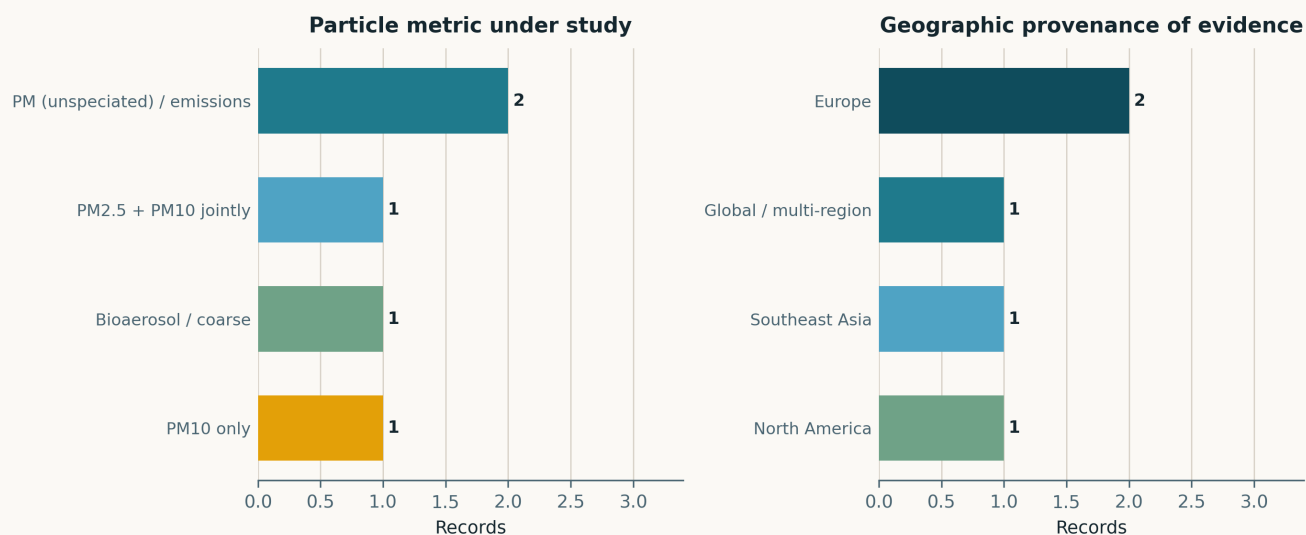


Fig. Left — particle metric: unspiciated PM in the two intervention records, PM₁₀ only in the dust survey (reported as a reservoir), bioaerosol in the aircraft campaign, and PM_{2.5} + PM₁₀ in the review. The dental study is the only record reporting a respirable (PM₄) cut. Right — geography: Europe two, and one each for North America, Southeast Asia and global / multi-region.

Life-course windows addressed today (records may span several stages)

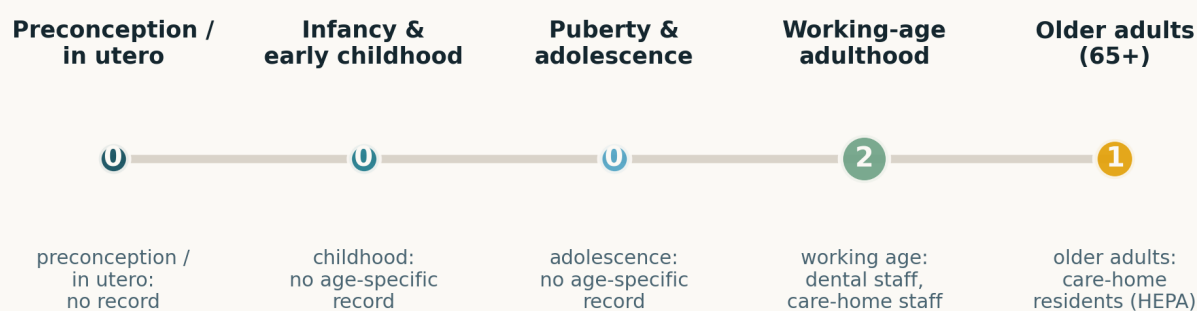


Fig. Life-course windows: working age two (dental providers, care-home staff) and older adults one (care-home residents under HEPA filtration). No record addresses in utero, childhood or adolescence — a consequence of a day dominated by occupational and environmental-survey work.

Where the work is happening: subtopic x study architecture



Fig. Subtopic × study architecture for the 5 records. Measurement campaigns spread across occupational, exposure and sensing; the two review-type records sit in occupational and mechanistic toxicology. Every occupied cell holds a single record except occupational measurement.

No forest plot this issue. No admitted abstract reports a PM effect estimate with a confidence interval. The dental study's estimand is a procedure-to-background concentration ratio with a 95 % lower confidence limit, reported per system in the full text rather than in the abstract; the AFRI-c qualitative arm reports no effect measure, and the remaining three records report none. The f5 panel is absent by design, not by build failure.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Mechanistic toxicology

1 record

Gao et al. 2026 Carcinogenesis

Air pollution and early-onset lung cancer: from environmental carcinogenesis to early intervention strategies

Narrative review of mechanisms linking air pollution to early-onset lung cancer: genomic instability (DNA damage, telomere dysfunction, mitochondrial impairment, mutational signatures), epigenetic dysregulation, a tumour-promoting niche through CXCL13 and IL-1beta, immune evasion via PD-L1 and CD47-SIRPalpha, and respiratory-gut microbiome disruption, closing with candidate biomarkers and chemopreventive targets. Limitation: no systematic search or evidence grading, the 99%-above-guideline and >14%-of-lung-cancer-deaths framing is quoted rather than re-derived, and the biomarkers are called 'validated' on the strength of single studies. Relevance: low as evidence, moderate as a map of which mediators a cohort with measured personal exposure could actually test.

doi: 10.1093/carcin/bgag070

Exposure assessment & modelling

1 record

Balabat et al. 2026 World J Microbiol Biotechnol

Urban Road Dust as a Microbial Hotspot: Spatial Variation of Bacterial Communities and Metal Associations in the Philippines

Road dust from nine sites across three metropolitan areas profiled by 16S rRNA metabarcoding with PICRUSt2 functional prediction and trace-metal pollution indices. Proteobacteria and Actinobacteria dominated everywhere with a persistent core of Paracoccus, Acinetobacter, Kocuria and Deinococcus; alpha diversity did not differ between cities (Kruskal-Wallis $p > 0.05$) but composition did (PERMANOVA $R^2 = 0.466$, $p = 0.004$). Paracoccus rose toward the most metal-laden Cagayan de Oro site and correlated with cobalt and nickel alongside raised predicted denitrification. Limitation: three sites per city, no replication within site, PICRUSt2 infers function rather than measuring it, and road dust is a reservoir - nothing here shows what becomes airborne or is inhaled. Relevance: low-moderate - the resuspended coarse fraction that low-cost optical nodes do see is biologically non-inert.

doi: 10.1007/s11274-026-05307-z

Sensing, forecasting & instrumentation

1 record

Metris and Metris 2026 PeerJ*Fungus above us: exploring the atmospheric mycobiome by aircraft*

Aircraft bioaerosol sampling across seasons, sequenced and joined to vegetation-productivity and meteorological data. Diversity fell with altitude and with increasing vegetation productivity; decomposers and pathogens dominated, with over 40% of taxa putative plant or animal pathogens (Neosascochyta, Cladosporium, Alternaria, Penicillium among the abundant genera); guild composition shifted with productivity and surface wind. Limitation: a small number of flight tracks stand in for a continent, relative abundance from amplicon data is not concentration, and viability is untested - 'pathogen' here is taxonomic assignment, not infectivity. Relevance: moderate - aircraft campaigns set the vertical context that ground bioaerosol sensors, which cannot resolve taxa at all, are implicitly assuming.

doi: 10.7717/peerj.21746

Occupational & indoor

2 records

Blackley et al. 2026 Ann Work Expo Health*Aerosol mitigation efficiency of high-volume suction systems during aerosol-generating dental procedures with patients*

Five commercial high-volume suction systems - standard HVE, three intraoral systems and one modified HVE - measured during real crown preparations (n=26) and ultrasonic scaling (n=25), with direct-reading instruments near the dentist and in the hallway. Procedure-to-background concentration ratios were modelled with covariates (suction flow rate, procedure duration, tooth location, scaler type and quadrant); a system counted as effective only when the 95% lower confidence limit of the ratio was ≤ 1 , i.e. indistinguishable from background. Limitation: unblinded single-site evaluation with modest per-system n, patients differ across procedures, and the short-duration ratio design cannot speak to a full clinical day. Relevance: high - this is the design a sensing group should copy. The estimand is 'did the control return the air to background', tested against the instrument's own noise, rather than a percentage reduction quoted without a comparison.

doi: 10.1093/annweh/wxag081

Rees et al. 2026 NIHR Open Res*Lessons learned from using high efficiency particulate air (HEPA) filtration units in care homes: interviews with care home staff, residents and relatives from the AFRI-c trial*

Reflexive thematic analysis of interviews with 25 staff, 20 residents and 12 relatives inside the AFRI-c pragmatic cluster RCT, whose main result was that HEPA units did not reduce respiratory infections. Units became normalised but were judged unsuitable for some residents; lockable settings and wall fixing were asked for; cost was seen as prohibitive sector-wide, raising an equity concern that filtration would reach only residents who can pay. Maintenance burden was minor. Limitation: qualitative, post hoc, and conducted among people who already knew the trial was running; no measurement of unit runtime, filter loading or achieved air-change rate - which is the missing variable that would explain a null. Relevance: high - a filtration null without exposure verification is the standing weakness of this trial literature, and a logging low-cost PM node in each room is the cheap fix.

doi: 10.3310/nihropenres.13993.1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Mechanistic toxicology					
Gao et al.	Air pollution and early-onset lung cancer: from environmental carcinogenesis to early intervention strategies	<i>Carcinogenesis</i>	Narrative review	Ambient and household PM _{2.5} /PM ₁₀ with PAH, metal and VOC components (review)	Global
Occupational & indoor					
Blackley et al.	Aerosol mitigation efficiency of high-volume suction systems during aerosol-generating dental procedures with patients	<i>Ann Work Expo Health</i>	Field measurement	Direct-reading number and mass concentrations, PM ₄ (respirable) and PM ₁₀ (thoracic), with size distribution	USA (NIOSH, clinical dental operatories)
Rees et al.	Lessons learned from using high efficiency particulate air (HEPA) filtration units in care homes: interviews with care home staff, residents and relatives from the AFRI-c trial	<i>NIHR Open Res</i>	Qualitative study (within RCT)	HEPA filtration units as the intervention; no PM measured in this component	United Kingdom (care homes, AFRI-c cluster RCT)
Sensing, forecasting & instrumentation					
Metris and Metris	Fungus above us: exploring the atmospheric mycobiome by aircraft	<i>PeerJ</i>	Field measurement	Airborne fungal bioaerosol, aircraft sampling 2022-2023 with remote-sensing and meteorological covariates	Europe (aircraft surveys; United Kingdom-based)
Exposure assessment & modelling					
Balabat et al.	Urban Road Dust as a Microbial Hotspot: Spatial Variation of Bacterial Communities and Metal Associations in the Philippines	<i>World J Microbiol Biotechnol</i>	Field measurement	Road dust (re-suspendable coarse reservoir): 16S V3-V4 metabarcoding, trace metals, pollution indices	Philippines (Metro Manila, Cebu, Cagayan de Oro)

Provenance and method

Entry window **4 October 2026**, single day, continuous with the 3 October issue. Harvest and screening for this day were completed on the **6 October** run and cached (`cache/2026-10-04/_fresh.json`); this run authored from that cache without re-querying. **13** fresh candidates after cross-day deduplication — a Saturday, and the lowest daily volume of the series. **5 admitted, 8 rejected** with logged reasons. **No record carried on metadata alone**, the first such issue since 27 September: the Crossref-ISSN leg returned nothing dated 4 October, and Elsevier does not deposit at weekends. **Consensus** returned 10 calibration hits, **0 new**. **Scholar Gateway** fails with `ACCESS_DENIED`. **ClinicalTrials.gov**: no PM or air-pollution registration updated in the window beyond EPIC-AIR (NCT07500948), carried from the 3 October issue. DOI gate: **0 fail, 0 warn**. Abstracts remain the copyright of their respective publishers.